

TARGET-90

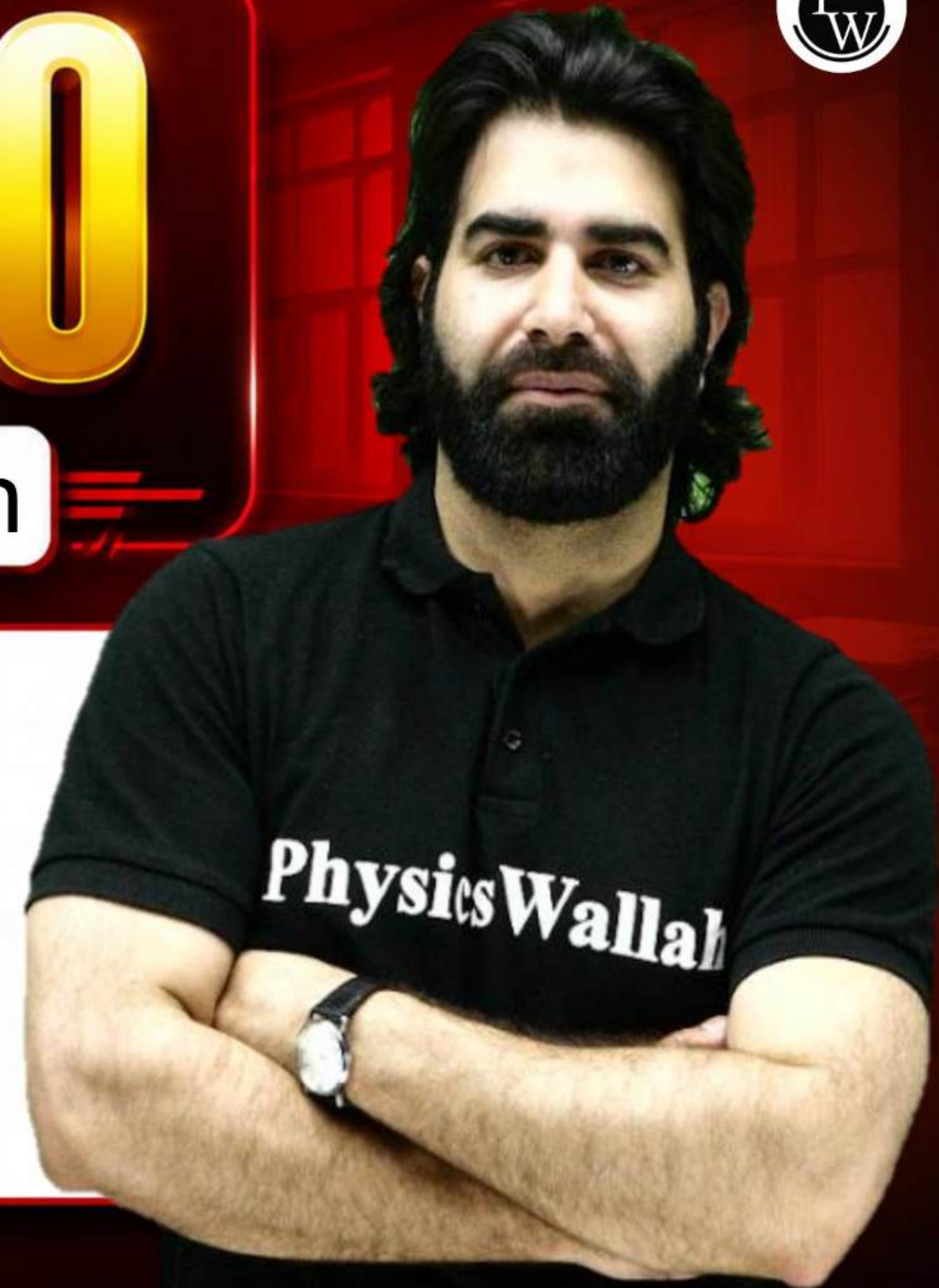
JKBOSE CLASS- 11th

Animal Kingdom

One Shot

Zoology

Adil sir



Topics To Be Covered

- 1 ... Basis of Classification.
- 2 ... Phylum . porifera → chordata
- 3 ...
- 4 ...

Animalia

(kingdom)

Animals

- Eukaryotic

- Multicellular

- w/o cell wall.

- Heterotrophic.

→ Divides

Phylum

↓
Class

↓
Order

↓
Family

↓
Genus

↓
Species

Main →

Porifera

Coelenterata

Ctenophora

Platyhelminths

Aschelminths

Annelida - Circulatory system

Arthropoda

Mollusca

Echinodermata

Hemichordata

Chordata

→ Cnidarians

→ Dimag me dal 10.

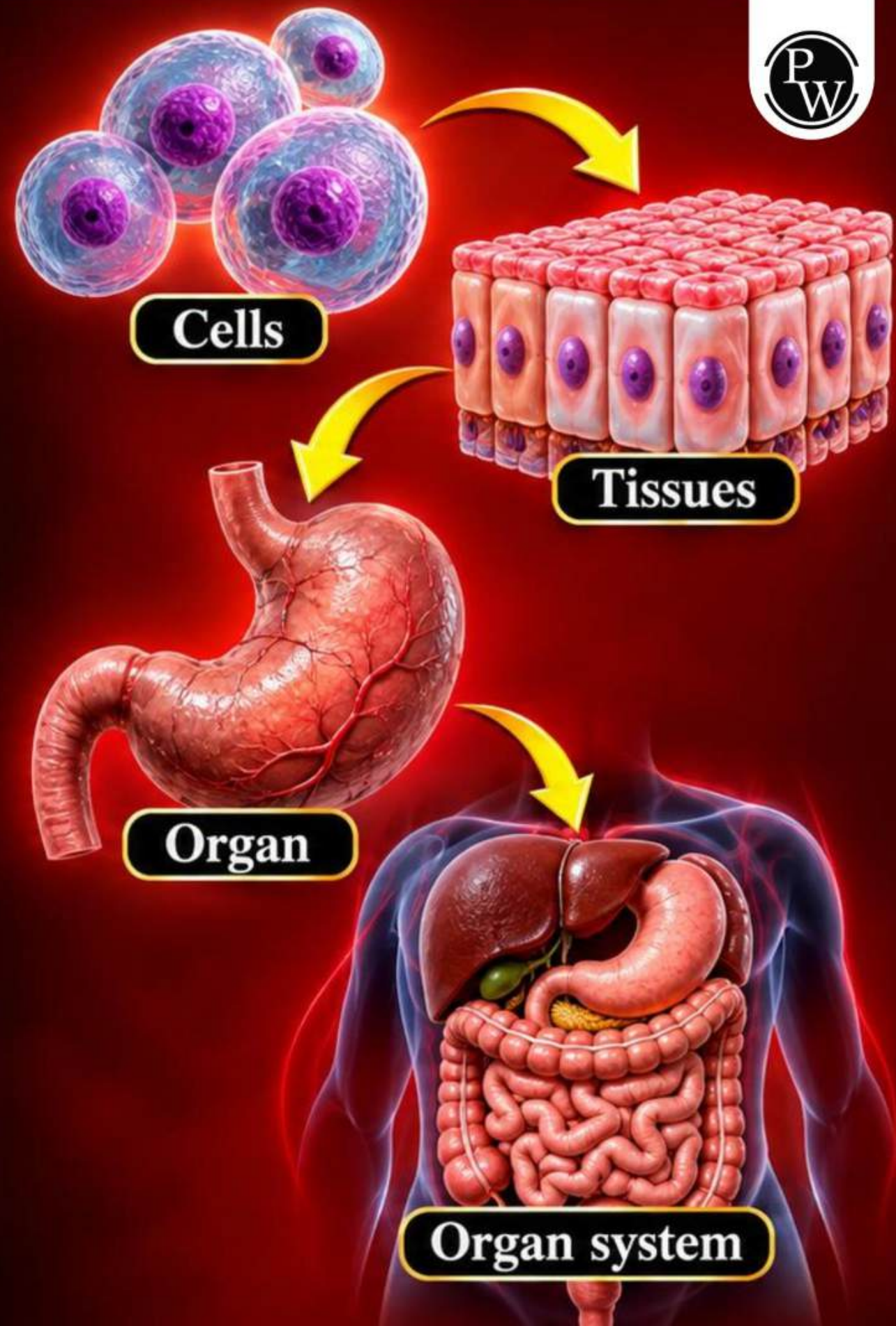
Flatworms

Nematohelminths
(Nematodes)

(Triblasts)

BASIS OF CLASSIFICATION

- Basis of classification are – Arrangement of cells, body symmetry, nature of coelom, patterns of digestive, circulatory and reproductive system.
- **(A) Level of Organization**
- Organisation is the structural differentiation of animal body.
- (a) Cellular level of organization: In this, the cells are arranged as loose cell aggregates.
Examples: Poriferans (Sponges). *① point*
- (b) Tissue level of organization: In this, the cells performing the same functions are arranged into tissues.
Examples: Coelenterates and Ctenophores.
- (c) Organ level of organization: In this type, the tissues are grouped together to form organs and the organs are associated to form organ systems. Each system performs a specific physiological function. Examples: All higher animals (from Platyhelminthes to chordates)

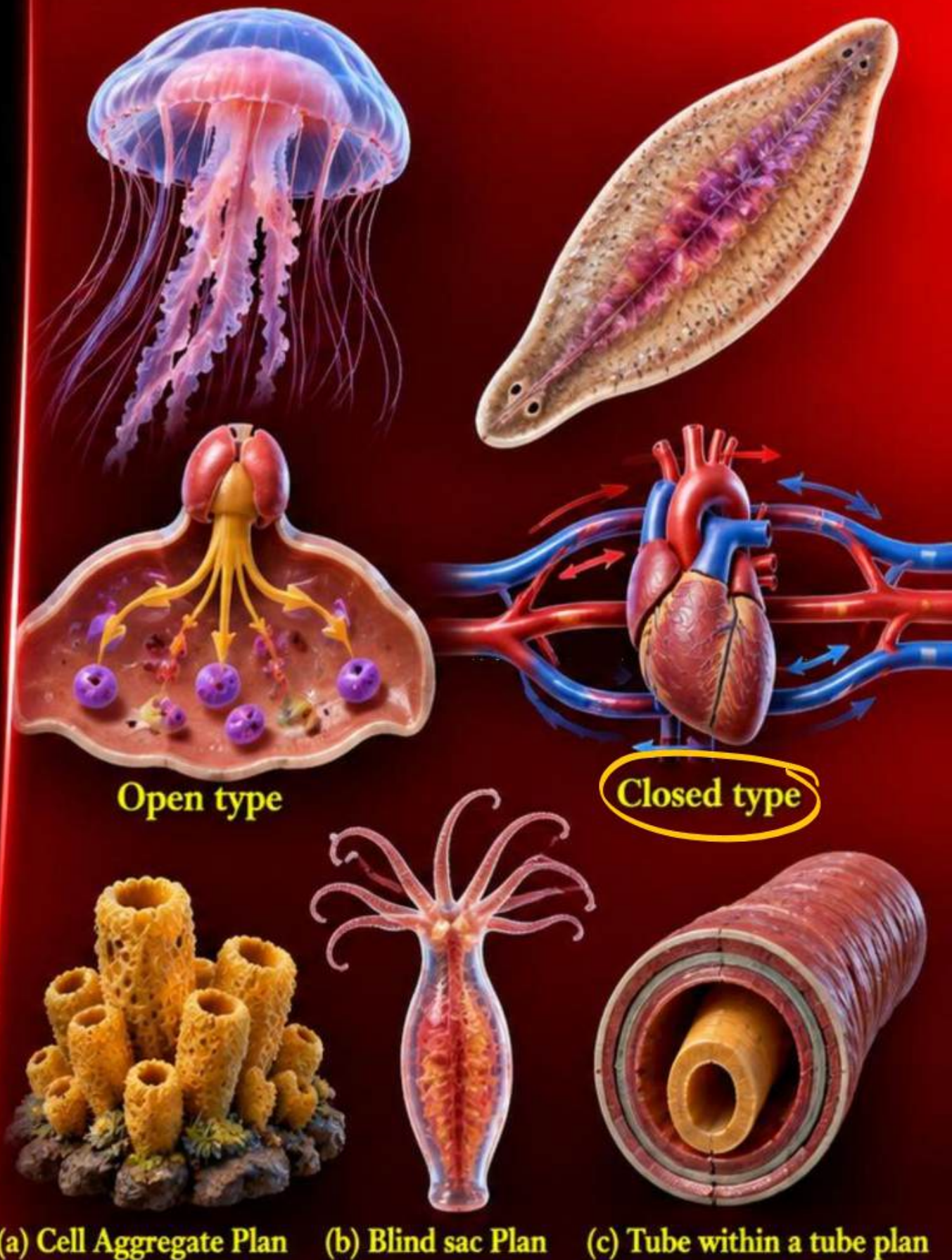


BASIS OF CLASSIFICATION

- Organ systems of different animals show complexities. Examples:
- **Digestive system** of ^{Cnidaria}cnidarians and Platyhelminthes is incomplete i.e. it has only a single opening which serves as both mouth and anus. Complete digestive system has 2 openings i.e. mouth and anus.
- **Circulatory system** is of 2 types: Open and closed.
 - In **open type**, blood is pumped out through heart. Cells and tissues are directly bathed in it. *Arthropoda, Mollusca, Hemichordata.*
 - In **closed type**: Blood is circulated through vessels. *→ Annelid, Chordata.*

2. Body Plan

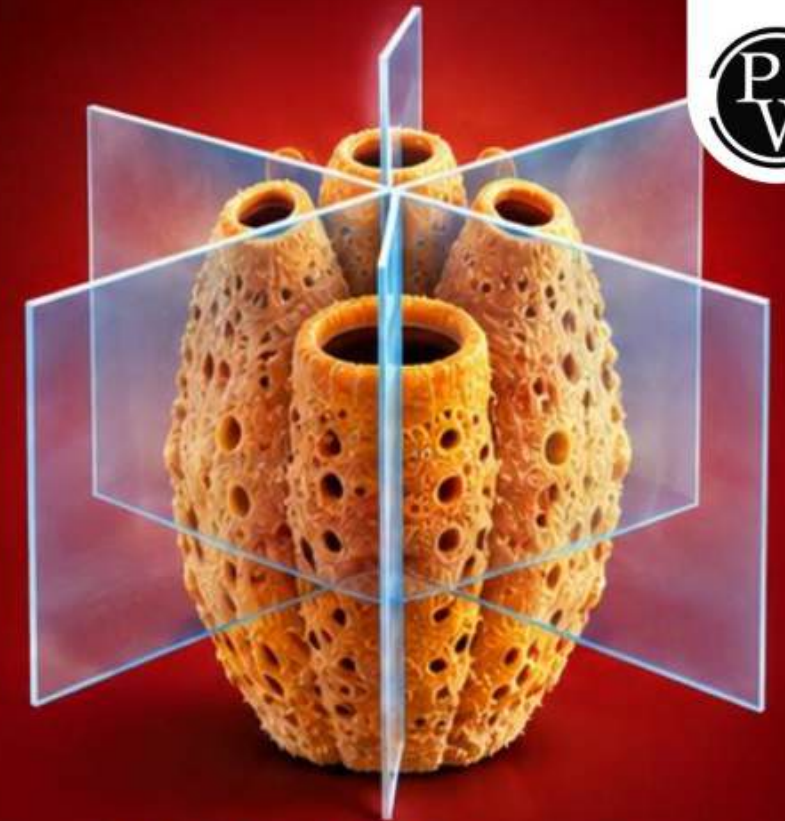
- Animals have three types of body plans.
- (a) **Cell Aggregate Plan**: In this, the body consists of aggregation of cell. It is usually found in Sponges.
- (b) **Blind sac Plan**: In this, the body has a single cavity with one opening to the outside. The single opening act as both mouth and anus.
- (c) **Tube within a tube plan**: In this, the body has two tubes; one formed by the body wall and other formed within it by the digestive tract.



(a) Cell Aggregate Plan (b) Blind sac Plan (c) Tube within a tube plan

BASIS OF CLASSIFICATION

- **3. Body Symmetry**
- Based on symmetry, animals are two types: Asymmetrical and Symmetrical.
- (a) **Asymmetrical**: In this, body cannot be divided into equal halves through median plane. The simplest animals have no symmetry. Examples: Sponges, Snails etc.
- (b) **Symmetrical**: In this, body can be divided into two similar parts. It is of two types.
 - (i) **Radial symmetry**: Body can be divided into 2 similar parts by any plane along oral aboral axis of body. Examples: Some Poriferans, Cnidarians, Ctenophores and Echinoderms.
 - (ii) **Bilateral symmetry**: Body can be divided into two identical right and left halves by a section passing through the longitudinal axis. Examples: Annelids, arthropods and all vertebrates.
- The body of bilaterally symmetrical animal has an upper or vertebral dorsal side, a lower ventral side, left and right lateral sides, anterior (cephalic) side and posterior (anal) side.



(a) Radial Symmetry

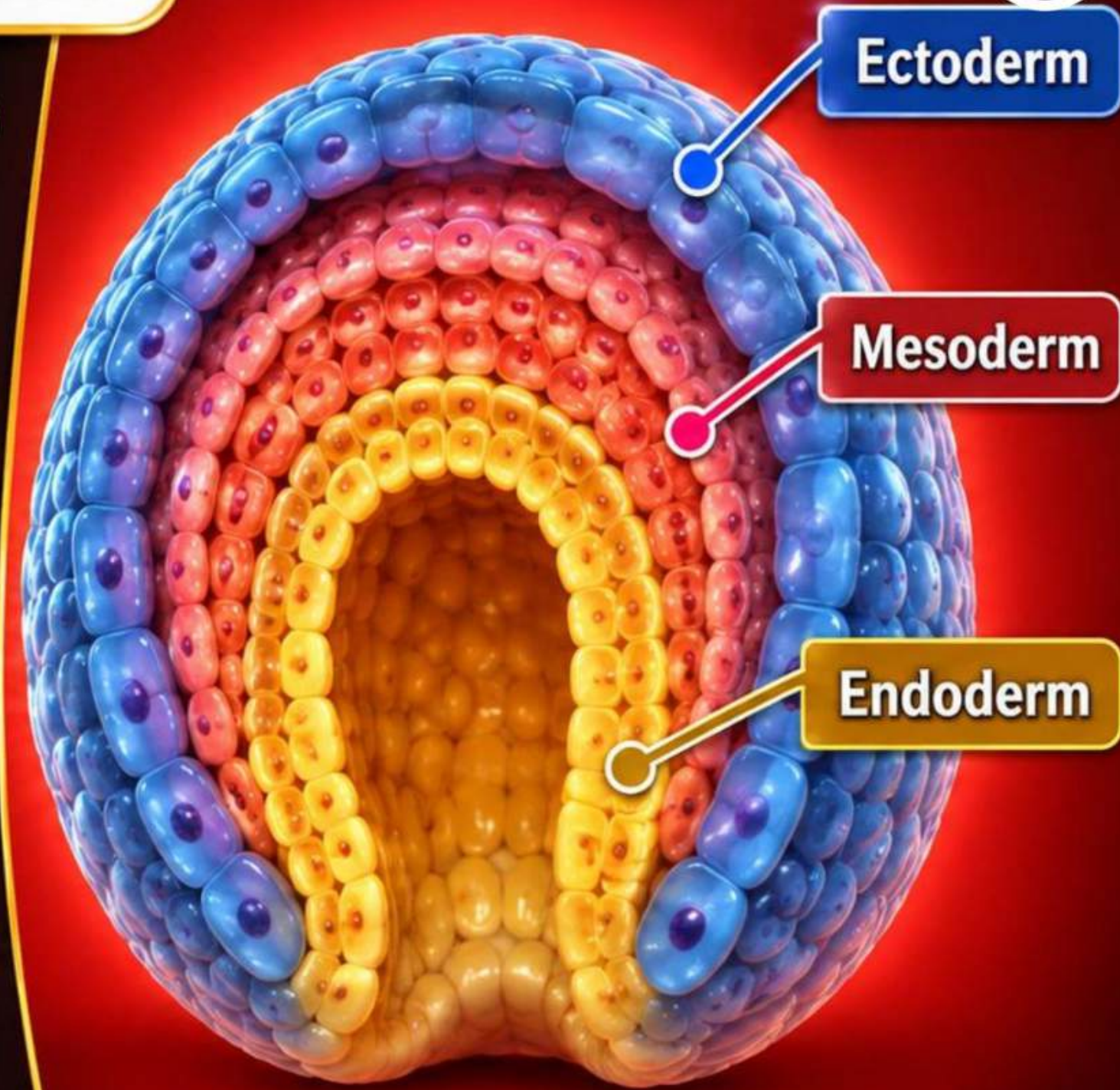


(b) Bilateral symmetry

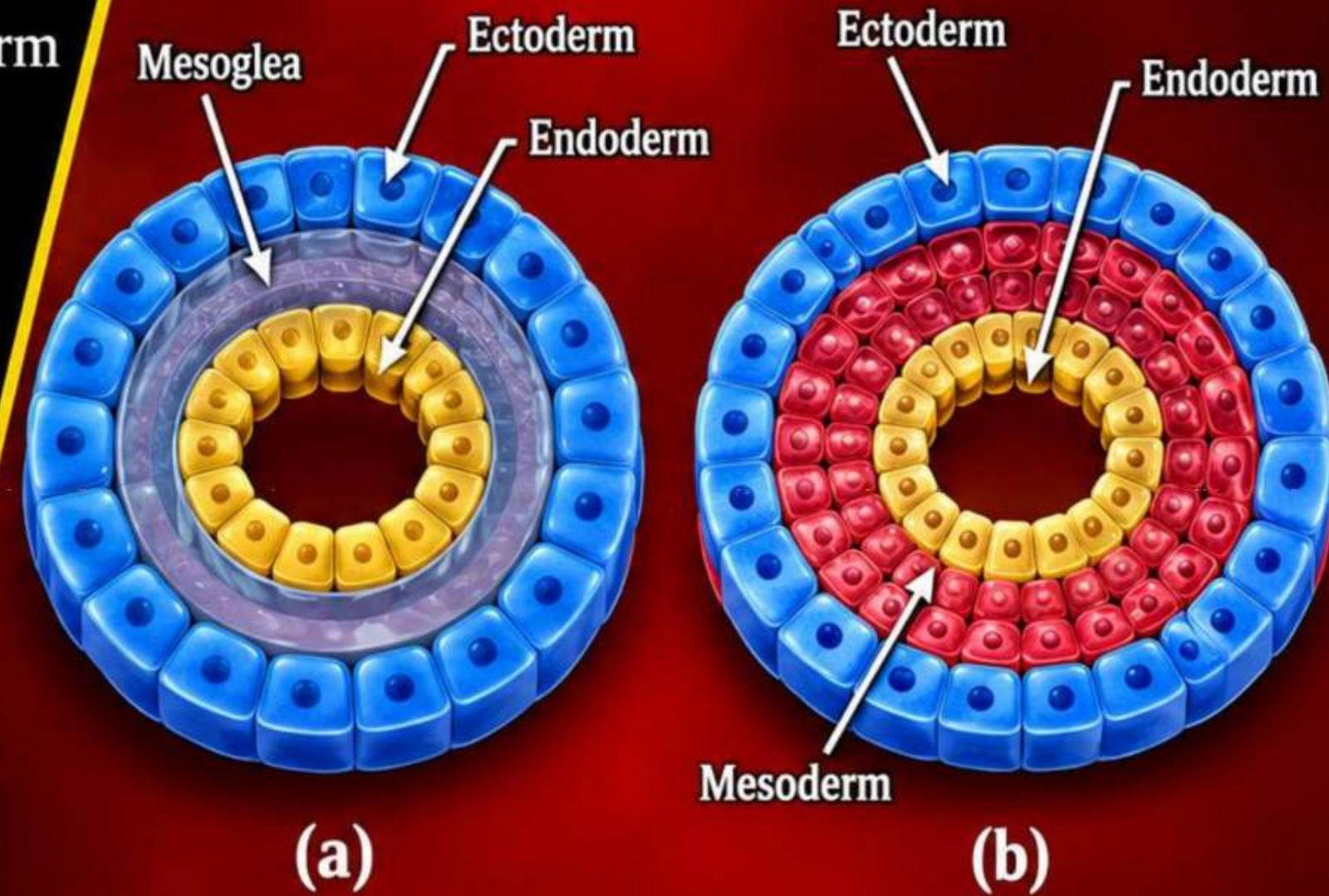
BASIS OF CLASSIFICATION

4. Germ layers

- These are layers of gastrula from which all the body organs are formed.
- All cells of the adult organism can be traced to one of the three germ layers. Some animals have only the inner and outer germ layers but more complex animals have mesoderm as well.
- Three types of germ layers:
 - (a) **Ectoderm** - It is the inner layer which gives rise to the outer covering of the animal (skin, hair, nails, feathers, and scales) and the nervous system.
 - (b) **Mesoderm** - It gives rise to muscles, skeleton, circulatory system, kidney, reproductive system
 - (c) **Endoderm** - It gives rise to digestive tract and associated organs.



- Based on the number of germ layers, animals are two types- Diploblastic and Triploblastic.
- **(i) Diploblastic animals:** Cells are arranged in two germ layers- outer ectoderm and inner endoderm.
- Mesoglea may be present in between ectoderm and endoderm
- Examples: Sponges and Coelenterates, *Ctenophora*.
- **(ii) Triploblastic animals:** They have three germ layers-
- Outer ectoderm, middle mesoderm and inner endoderm.
- Examples: Platyhelminthes (Flat worms) to Chordates (mammals).

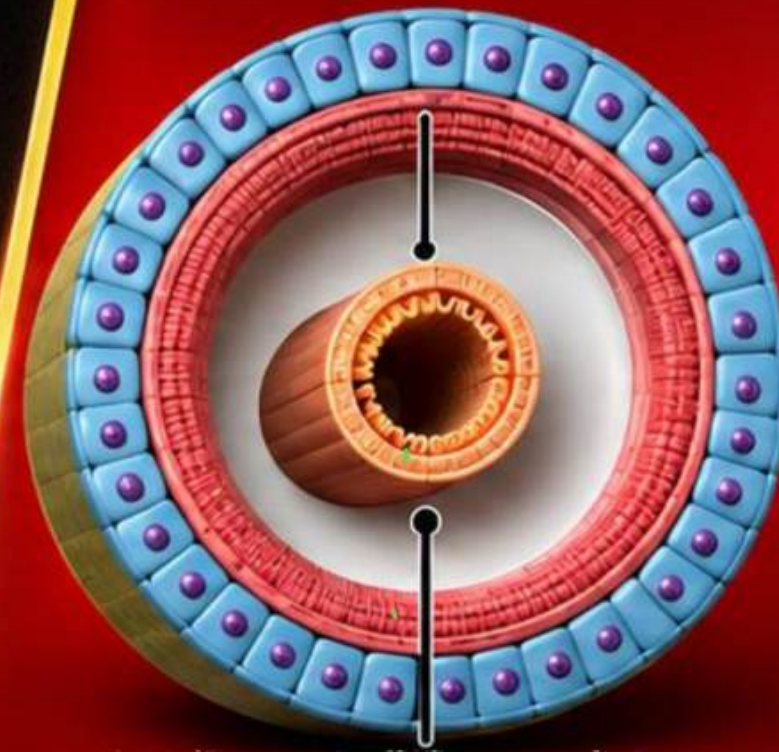


ig. Germinal layers: (a) Diploblastic (b) Triploblastic

5. Coelom (Body cavity)

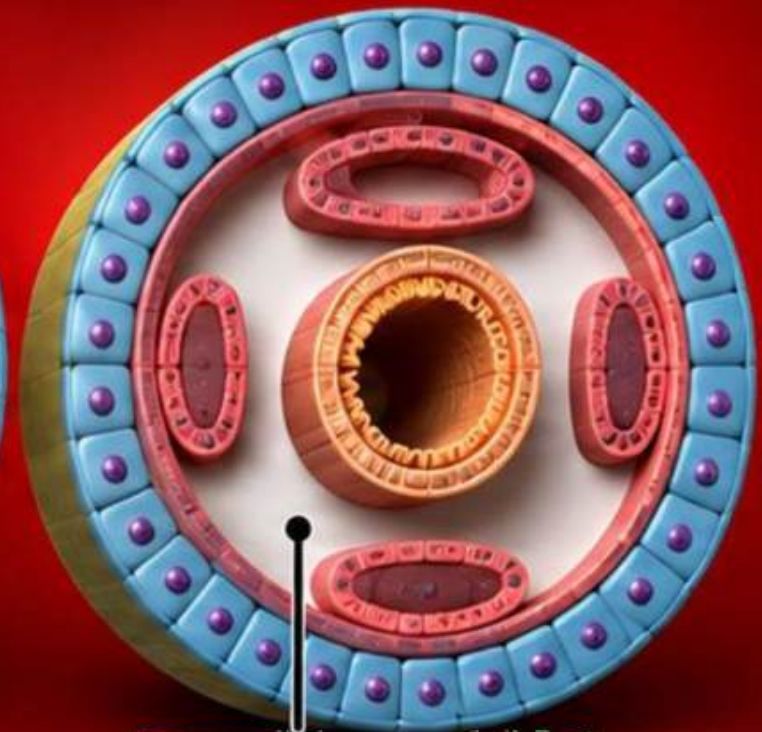
- It is the space between body wall and gut wall.
- Coelom separates the muscles of gut and body wall.
- On the basis of nature of coelom, animals are of 3 types: Acoelomate, Pseudocoelomate, and Eucoelomate.
- (a) **Acoelomate**: They have no coelom. The space between body wall and digestive cavity is filled with matrix (parenchyma). Examples: Poriferans to Platyhelminthes.
- (b) **Pseudocoelomate**: They have no true coelom. They have a body cavity partially surrounded by mesoderm. Mesoderm is present in scattered pouches between ectoderm and endoderm. Example: Aschelminthes.
- (c) **Coelomate (True coelomate or Eucoelomate)**: They have body cavity completely surrounded by mesoderm. Coelom is lined by peritoneal layer and is filled with coelomic fluid. Examples: Annelids to chordates.

(a) Coelomate



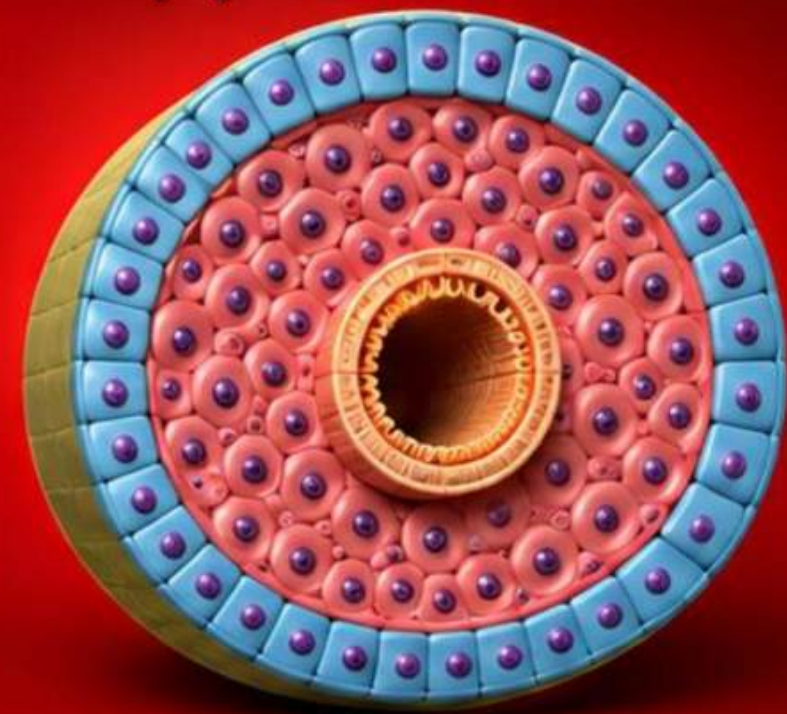
Coelom

(b) Pseudocoelomate



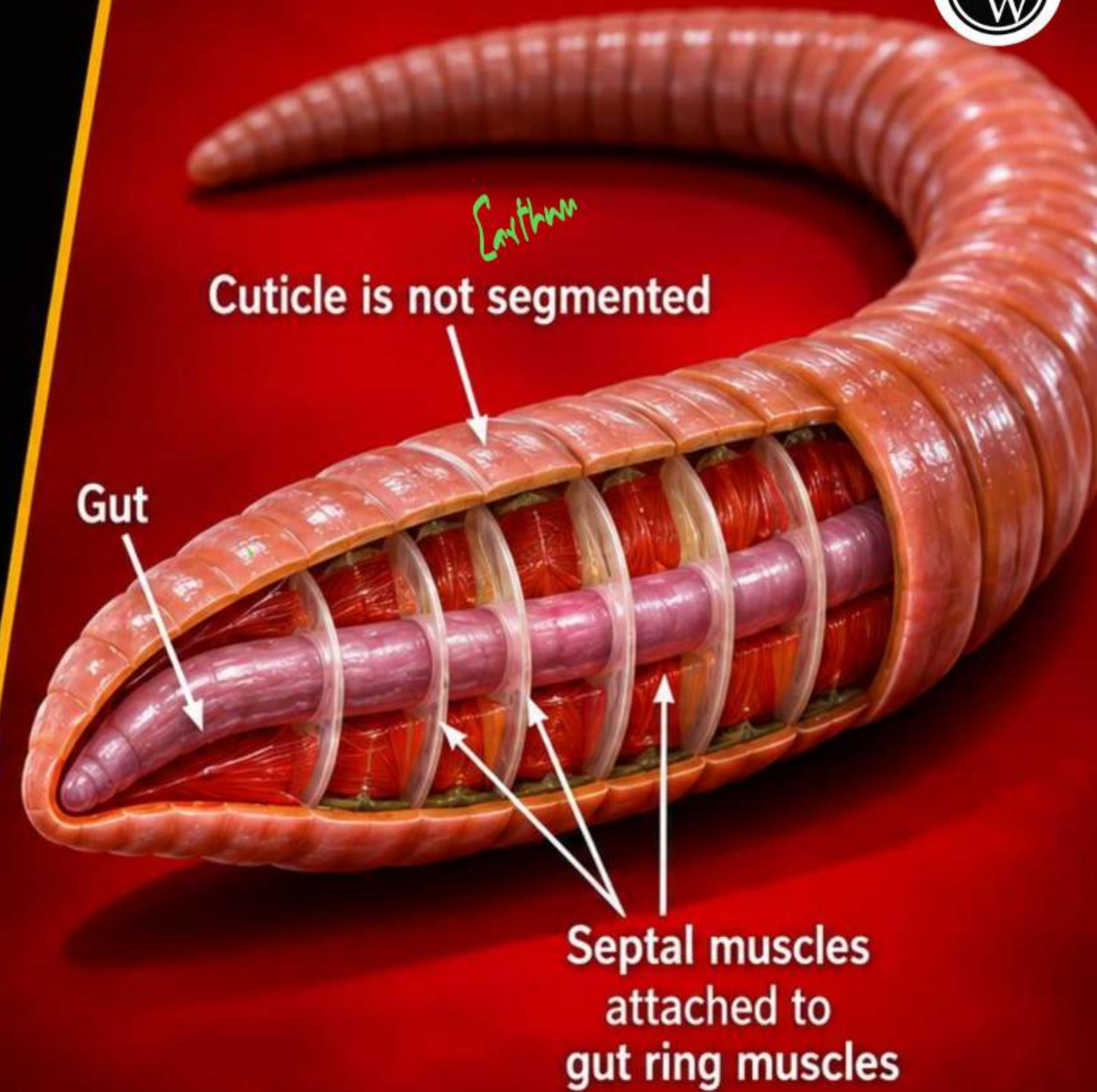
Pseudocoelom

(c) Acoelomate



● 6. Segmentation (Also known as Metamerism)

- Segmentation is the division or differentiation of the body into distinct proportions called segments.
- Examples: Annelids, Arthropods, In vertebrates: only internal metamerism is seen.
 - Chordata.
- It is of two types:
- (a) **Metameric segmentation (True metamerism):**
In this, the body is often divided both externally and internally into number of segments (metameres). E.g. Annelids, arthropods and chordates.
- (b) **Pseudometamerism (False metamerism):**
In this, the body is not internally divided. For e.g. the proglottids (segments of tapeworms) are budded off from the neck and are not of embryonic origin.
 → Tapeworm.

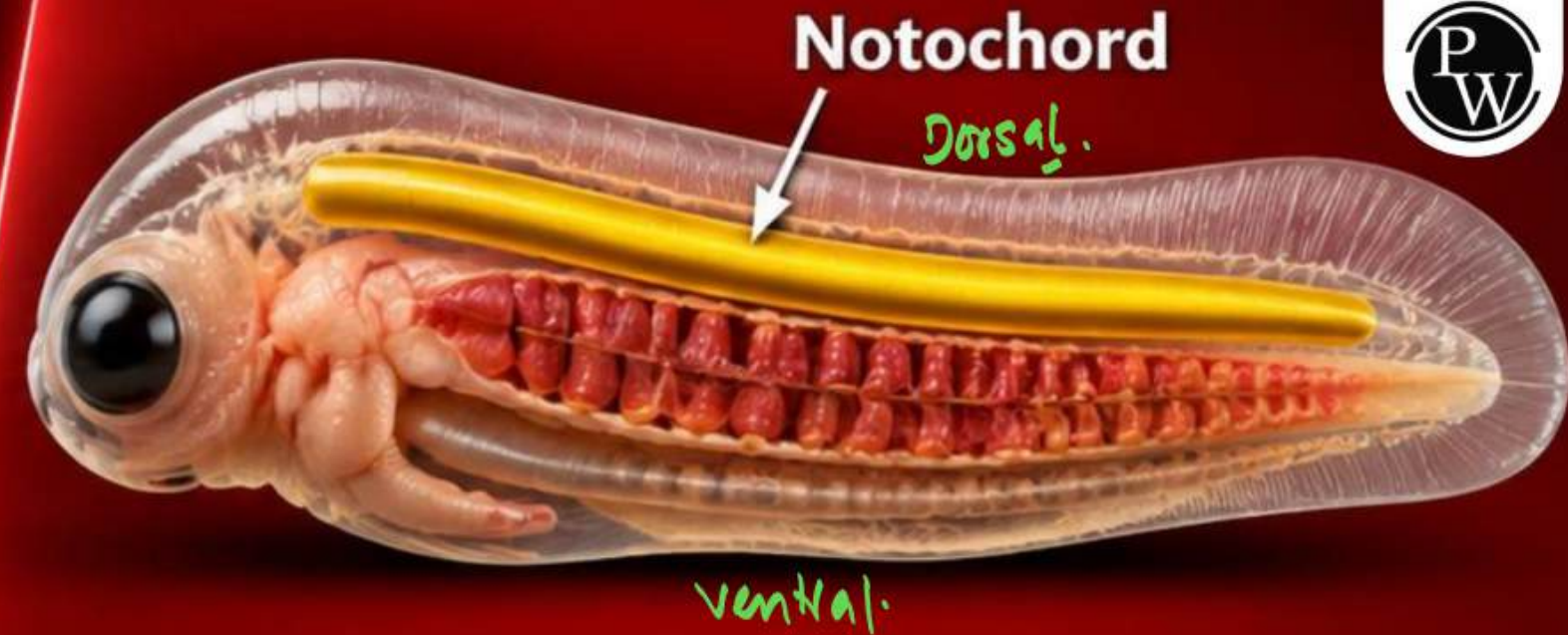


7. Notochord

- Notochord is a rod-like structure formed during embryonic development on the dorsal side.
- It is mesodermally derived.
- Animals with notochord are called **chordates** while those without notochord are called **non-chordates**.

8. Digestive tract

- It is the passage where food is taken for digestion, absorption and elimination. It is of two types:
- (a) **Incomplete digestive tract:** It has a single opening called mouth that takes food as well as eliminates the undigested food.
- It is found in Cnidarians and Platyhelminthes.
- (b) **Complete digestive tract:** It has two openings; mouth for intake of food and anus for elimination of undigested food.
- It is found in Nematelminthes to chordates.



Incomplete digestive tract



Complete digestive tract

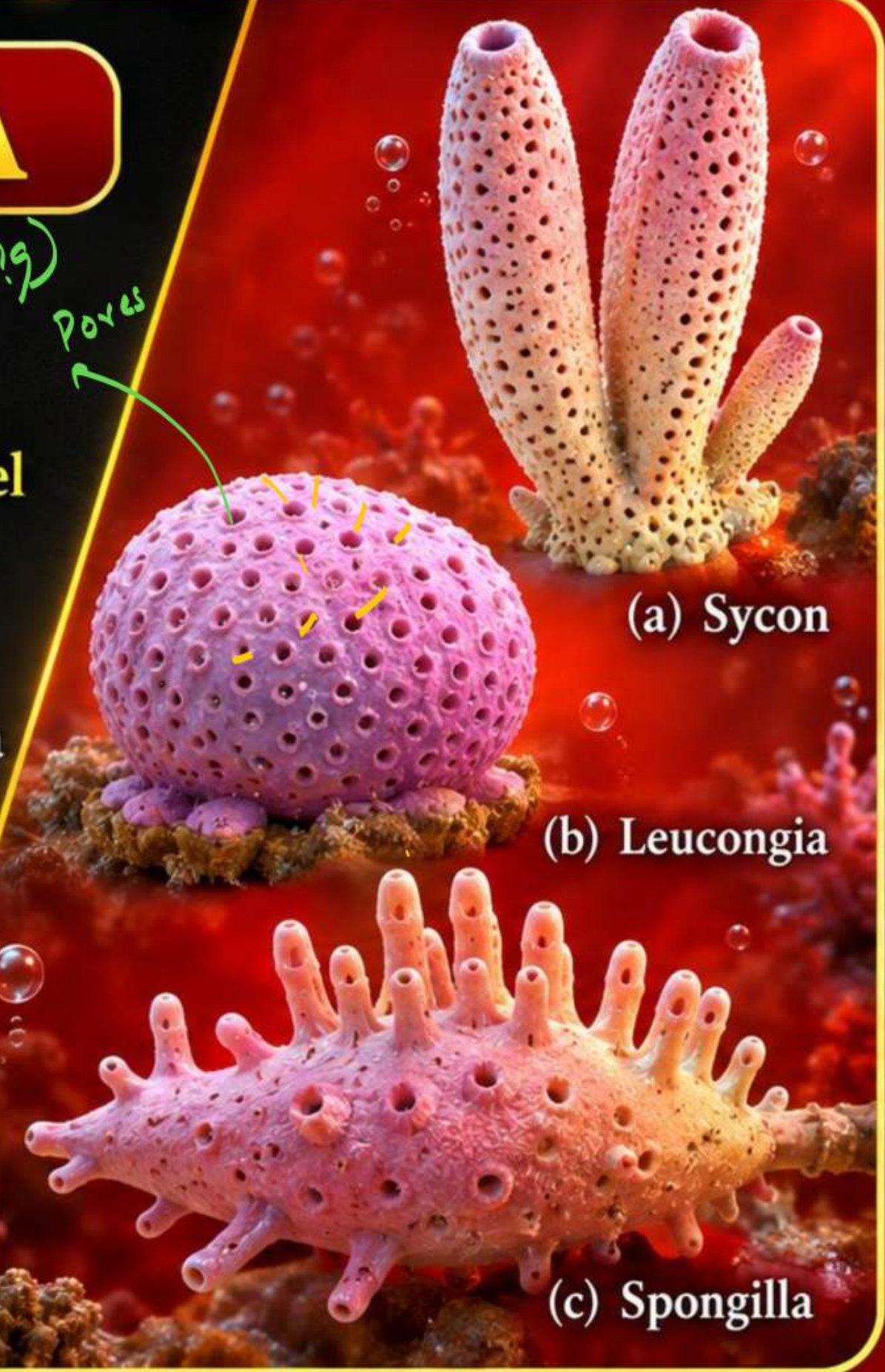


PHYLUM PORIFERA

- They are commonly known as **sponges**.
- They are generally **marine** and **mostly asymmetrical** animals.
- These are **primitive multicellular** animals and have **cellular level** of organisation.
- Sponges have a **water transport** or canal system.
- Water enters through minute **pores (ostia)** in the **body wall** into a **central cavity, spongocoel** from where it goes out through the **osculum**.
- This pathway of water transport is helpful in **food gathering, respiratory exchange** and **removal of waste**.
- **Choanocytes** or collar cells line the spongocoel and the canals.
- Digestion is **intracellular**.

Means (Pore Bearing)

Pores

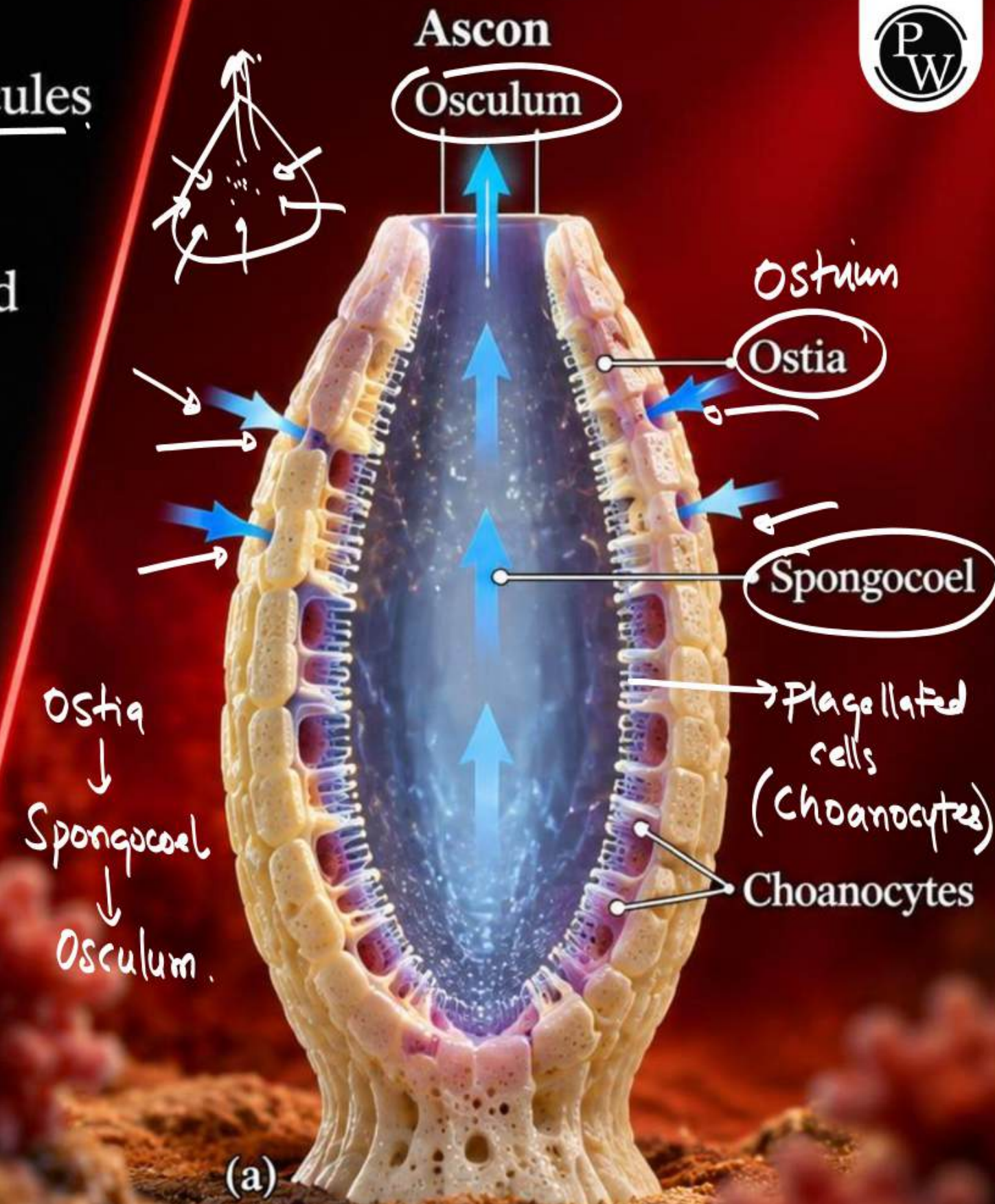


(a) Sycon

(b) Leucongia

(c) Spongilla

- The body is supported by a skeleton made up of spicules or spongin fibres.
- Sexes are not separate (hermaphrodite), i.e., eggs and sperms are produced by the same individual.
- Sponges reproduce asexually by fragmentation and sexually by formation of gametes.
- Fertilisation is internal. ✓✓
- Development is indirect having a larval stage which is morphologically distinct from the adult.
- Examples: Sycon (Scypha), Spongilla (Fresh water sponge) and Euspongia (Bath sponge).



Phylum – Coelenterata (Cnidaria)

- They are aquatic, mostly marine,
- Sessile or free-swimming,
- Radially symmetrical animals
- The name cnidaria is derived from the cnidoblasts or cnidocytes (which contain the stinging capsules or nematocytes) present on the tentacles and the body.
- Cnidoblasts are used for anchorage, defense and for the capture of prey.
- Tissue level of organisation and are diploblastic.
- They have a central (gastro-vascular) cavity with a single opening, hypostome.
- Digestion is extracellular and intracellular.
- Corals have a skeleton composed of calcium carbonate.
- Exhibit two basic body forms called polyp and medusa.

Cavity
Coelenteron

Sting cell
(Cnidoblast)

Tentacles

Coelenteron



Cnidoblast
(Cnidocyte)

Jelly fish

- Umbrella
- Free swimming
- Sexual

- Cylindrical
- Sessile
- Asexual

Polyp

Medusa

- Polyp is a sessile and cylindrical form like Hydra, Adamsia, etc.
- Medusa is umbrella-shaped and free-swimming like Aurelia or jelly fish.
- Those cnidarians which exist in both forms exhibit alternation of generation (Metagenesis), i.e., polyps produce medusae asexually and medusae form the polyps sexually (e.g., Obelia).
- Examples: Physalia (Portuguese man-of-war), Adamsia (Sea anemone), Pennatula (Sea-pen), Gorgonia (Sea-fan) and Meandrina (Brain coral).



(a) Aurelia (Medusa)



(b) Physalia



(b) Adamsia (Polyp)



(d) Pennatula



(e) Sea fan



Phylum – Ctenophora



- Ctenophores, commonly known as sea walnuts or comb jellies.
- Exclusively marine,
- Radially symmetrical,
- Diploblastic organisms with tissue level of organisation.
- The body bears eight external rows of ciliated comb plates, which help in locomotion.
- Digestion is both extracellular and intracellular.
- Bioluminescence (the property of a living organism to emit light) is well-marked in ctenophores.
- Sexes are not separate. (Hermaphrodites).
- Reproduction takes place only by sexual means.
- Fertilisation is external with indirect development.
- Examples: Pleurobrachia and Ctenoplana.

Comb plates
Jelly like
Comb plate

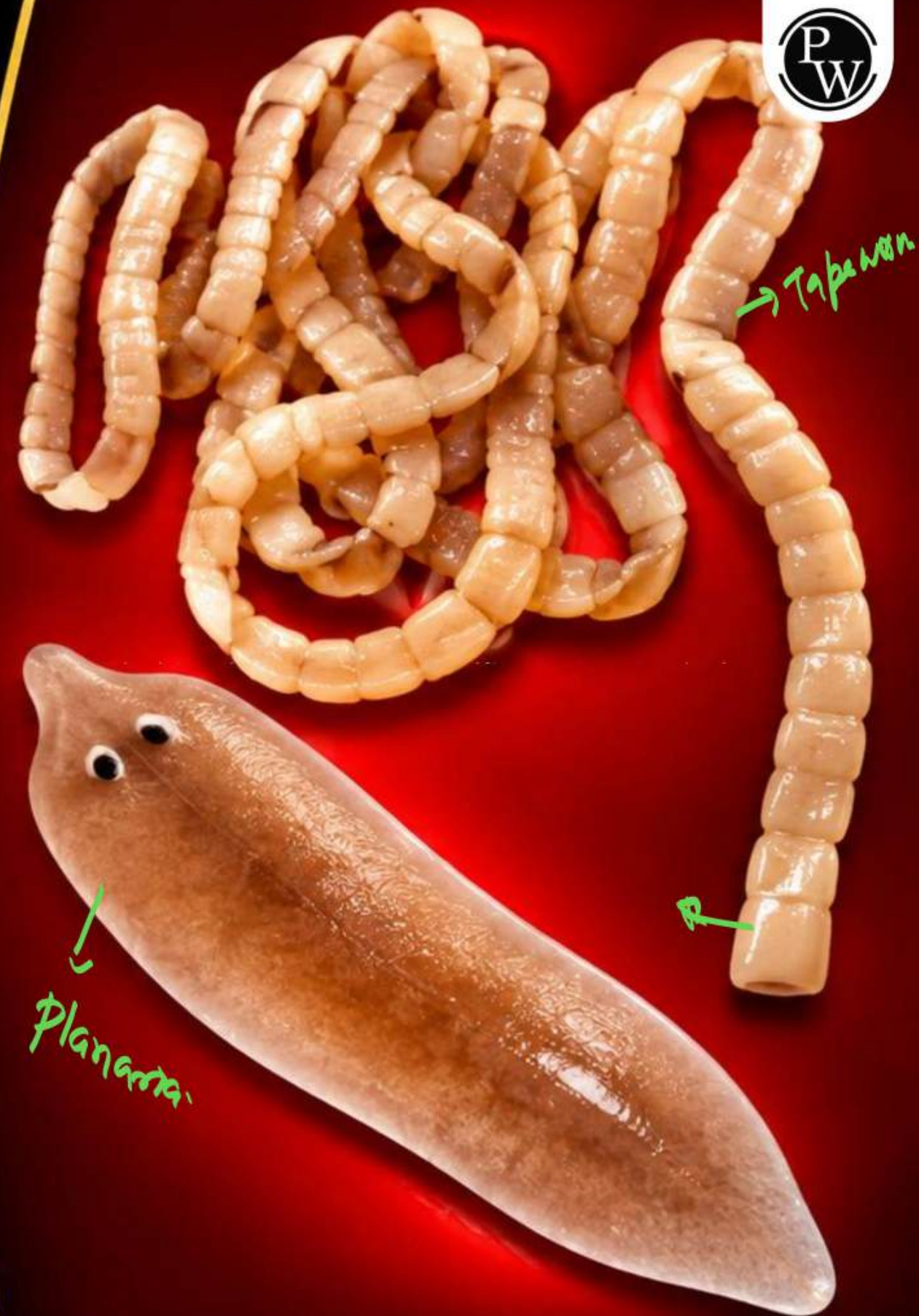
Cilia



Figure 4.8 Example of Ctenophora (*Pleurobrachia*)

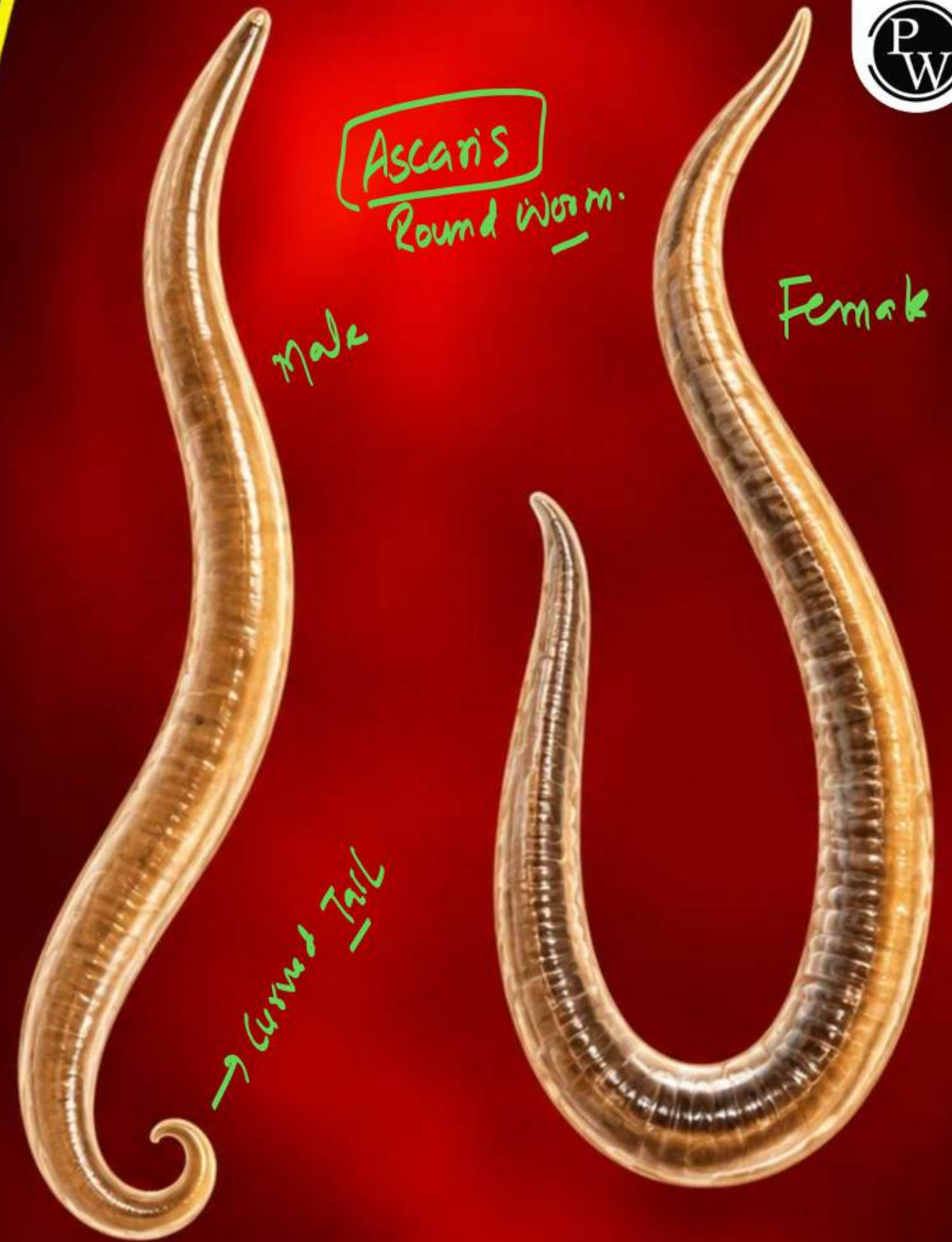
Phylum – Platyhelminthes

- They have dorso-ventrally flattened body, hence are called flatworms.
- These are mostly endoparasites found in animals including human beings.
- Bilaterally symmetrical, triploblastic → Ecto
Meso
Endoderm.
- Acoelomate animals with organ level of organisation.
- Hooks and suckers are present in the parasitic forms.
- Some of them absorb nutrients from the host directly through their body surface.
- Specialised cells called flame cells help in osmoregulation and excretion.
- Sexes are not separate → Protonephridia
(Hermaphrodite)
- Fertilisation is internal and development is through many larval stages.
- Planaria possess high regeneration capacity.
- Examples: Taenia (Tapeworm), Fasciola (Liver fluke).



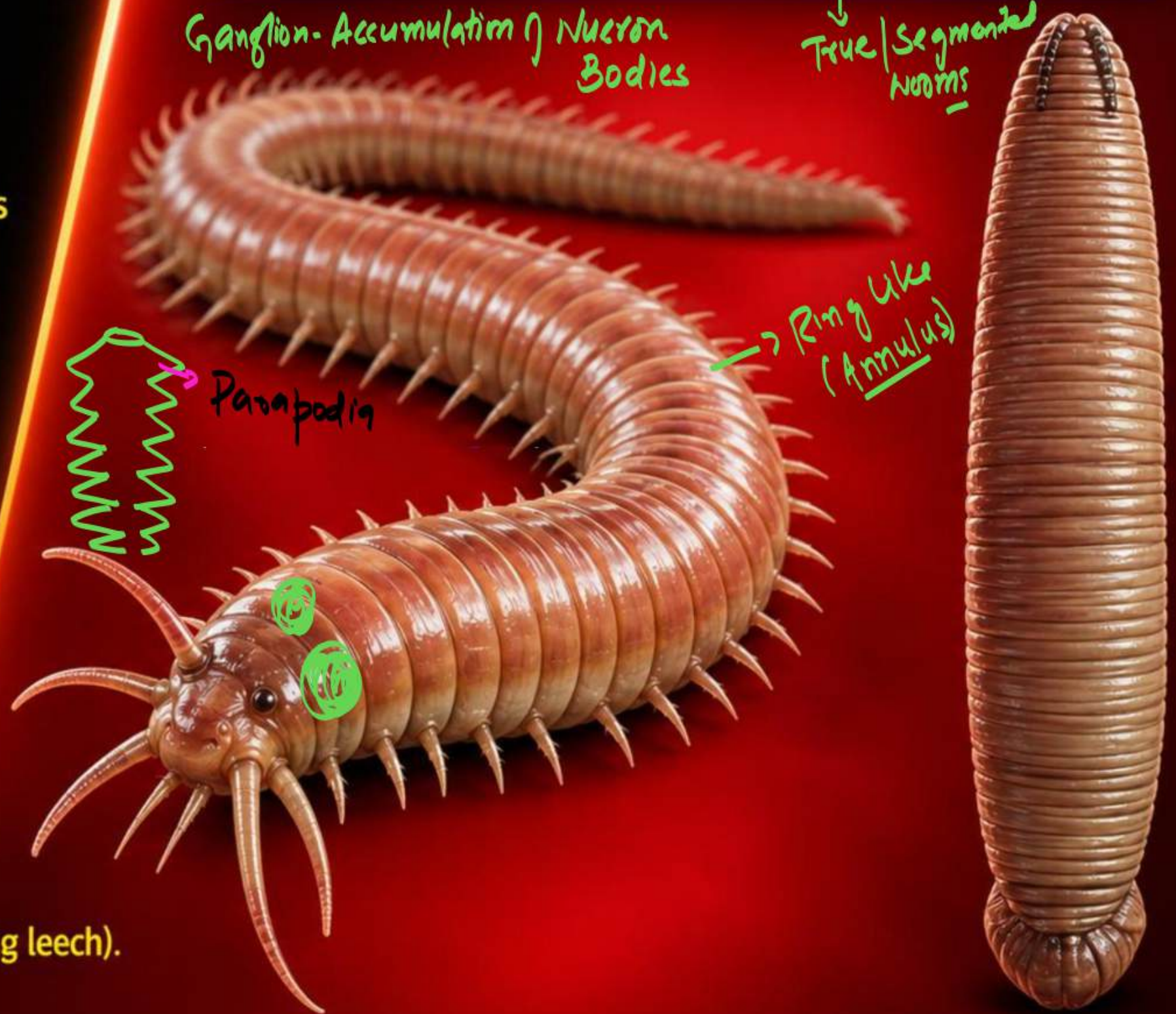
Phylum – Aschelminthes

- The body of the aschelminthes is circular in cross-section, hence, the name roundworms.
- They may be freeliving.
- aquatic and terrestrial
- parasitic in plants and animals.
- organ-system level of body organisation.
- They are bilaterally symmetrical, triploblastic and pseudocoelomate animals.
- Alimentary canal is complete with a welldeveloped muscular pharynx. *Two Opening*
- An excretory tube removes body wastes from the body cavity through the excretory pore.
- Sexes are separate (dioecious), i.e., males and females are distinct.
- Often females are longer than males.
- Fertilisation is internal and development may be direct (the young ones resemble the adult) or indirect.
- Examples : Ascaris (Round Worm), Wuchereria (Filaria worm), Ancylostoma (Hookworm).



Phylum – Annelida

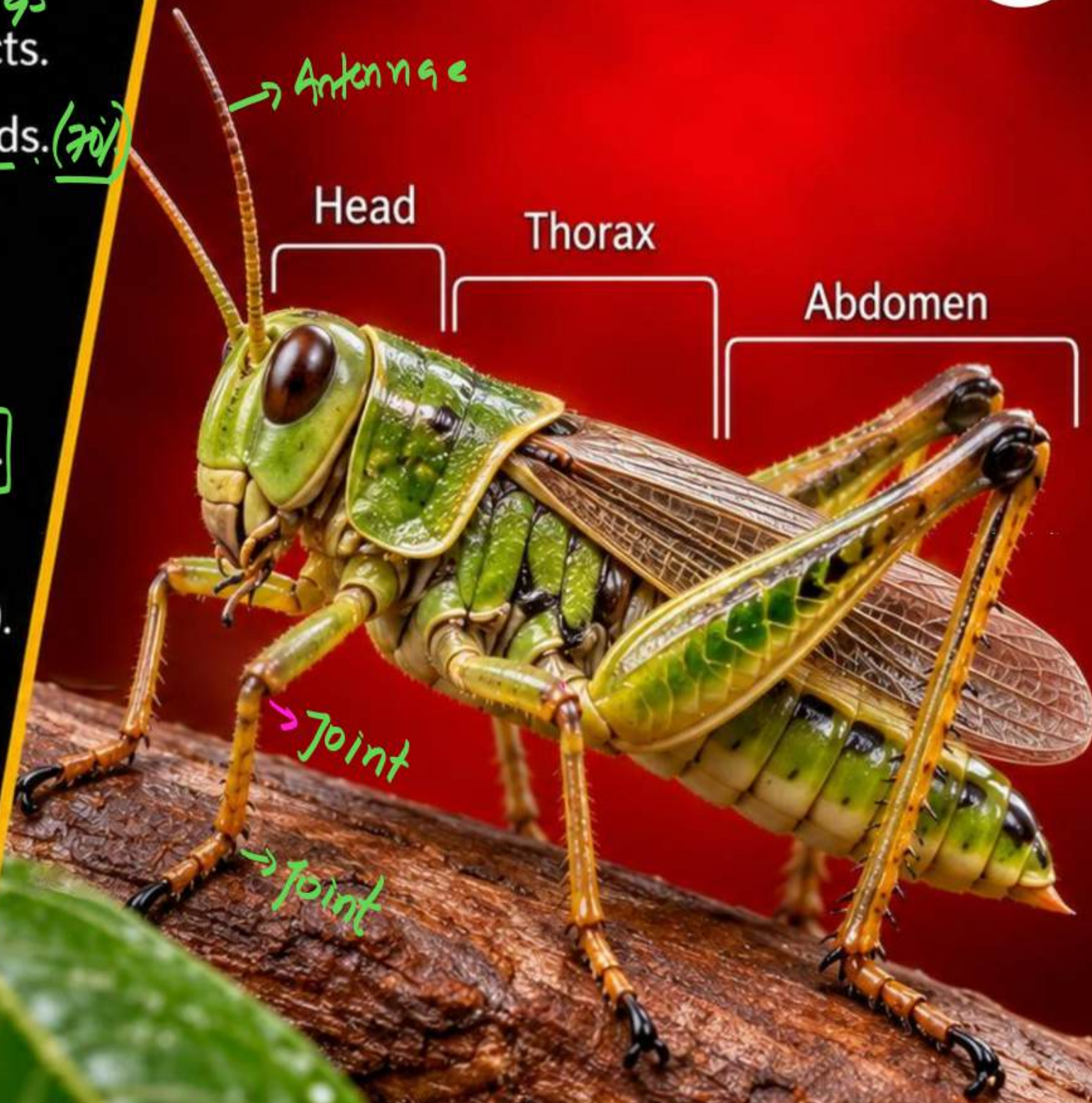
- They may be aquatic (marine and fresh water) or terrestrial; free-living, and sometimes parasitic.
- They exhibit organ-system level of body organization bilateral symmetry.
- They are triploblastic, metamerically segmented (True segmentation) coelomate animals.
- Their body surface is distinctly marked out into segments or metameres (Latin, annulus : little ring) and, hence, the phylum name Annelida.
- They possess longitudinal and circular muscles which help in locomotion.
- Aquatic annelids like Nereis possess lateral appendages, parapodia, which help in swimming.
- A closed circulatory system is present.
- Nephridia (sing, nephridium) help in osmoregulation and excretion.
- Neural system consists of paired ganglia (sing, ganglion) connected by lateral nerves to a double ventral nerve cord.
- Nereis, an aquatic form, is dioecious, but earthworms and leeches are monoecious. (Hermaphrodite)
- Reproduction is sexual.
- Nereis, Pheretima (Earthworm) and Hirudinaria (Blood sucking leech).



Phylum – Arthropoda



- This is the largest phylum of Animalia which includes insects.
- Over two-thirds of all named species on earth are arthropods. (70%)
- They have organ-system level of organisation.
- They are bilaterally symmetrical, triploblastic, segmented and coelomate animals.
- The body of arthropods is covered by chitinous exoskeleton. *Carbohydrate*
- The body consists of head, thorax and abdomen.
- They have jointed appendages (arthros-joint, poda-appendages).
- Respiratory organs are gills, book gills, book lungs or tracheal system.
- Circulatory system is of open type.
- Sensory organs like antennae eyes (compound and simple), statocysts or balance organs are present.
- Excretion takes place through malpighian tubules.



- They are mostly dioecious. (Sexes are separate)
- Fertilisation is usually internal.
- They are mostly oviparous. (Egg laying)
- Development may be direct or indirect.
- Examples:
- Economically important insects –
Apis (Honey bee),
Bombyx (Silkworm),
Laccifer (Lac insect)
- Vectors – Anopheles, Culex and Aedes (Mosquitoes)
- Gregarious pest – Locusta (Locust)
- Living fossil – Limulus (King crab).



Phylum – Mollusca

- This is the **second largest** animal phylum.
- Molluscs are **terrestrial** or **aquatic** (marine or fresh water).
- **organ-system** level of organisation.
- They are **bilaterally symmetrical**, **triploblastic** and **coelomate** animals.
- Body is covered by a **calcareous shell** and is **unsegmented**.
- **distinct head**, **muscular foot** and **visceral hump**.
- A soft and spongy layer of skin forms a **mantle** over the visceral hump.
- The space between the hump and the mantle is called the **mantle cavity** in which feather like gills are present.
- They have **respiratory** and **excretory** functions.
- The anterior head region has **sensory tentacles**.
- The mouth contains a file-like rasping organ for feeding, called **radula**.
- They are usually **dioecious** and **oviparous** with indirect development.



Phylum – Mollusca



- Examples:

- Pila (Apple snail),

- Pinctada (Pearl oyster),

- Sepia (Cuttlefish),

- Loligo (Squid),

- Octopus (Devil fish),

- Aplysia (Seahare),

- Dentalium (Tusk shell)

- Chaetopleura (Chiton).



Phylum – Echinodermata

- These animals have an endoskeleton of calcareous ossicles (small bones) and, hence, the name Echinodermata (Spiny bodied).
- All are marine with organ-system level of organisation.
- The adult echinoderms are radially symmetrical but larvae are bilaterally symmetrical.
- They are triploblastic and coelomate animals.
- Digestive system is complete with mouth on the lower (ventral) side and anus on the upper (dorsal) side.
- The most distinctive feature of echinoderms is the presence of water vascular system which helps in locomotion, capture and transport of food and respiration.
- An excretory system is absent.
- Sexes are separate.
- Reproduction is sexual.
- Fertilisation is usually external.
- Development is indirect with free-swimming larva.



- **Asterias** (Star fish),
- **Echinus** (Sea urchin),
- **Antedon** (Sea lily),
- **Cucumaria** (Sea cucumber)
- **Ophiura** (Brittle star).



Phylum – Hemichordata

- Hemichordata was earlier considered as a sub-phylum under phylum Chordata. (Chordata) ↓ Not chord
- But now it is placed as a separate phylum under non-chordata. (But Actually Stomach chord)
- worm-like marine animals
- organ-system level of organisation.
- They are bilaterally symmetrical, triploblastic and coelomate animals.
- The body is cylindrical and is composed of an anterior proboscis, a collar and a long trunk
- Circulatory system is of open type.
- Respiration takes place through gills.
- Excretory organ is proboscis gland.
- Sexes are separate.
- Fertilisation is external.
- Development is indirect.
- Examples: Balanoglossus and Saccoglossus. (Tongue worm)



Phylum – Chordata

- Animals belonging to phylum Chordata are fundamentally characterised by the presence of a notochord, a dorsal hollow nerve cord and paired pharyngeal gill slits. + Post Anal tail.
- These are bilaterally symmetrical, triploblastic, coelomate with organ-system level of organisation.
- They possess a post anal tail and a closed circulatory system.

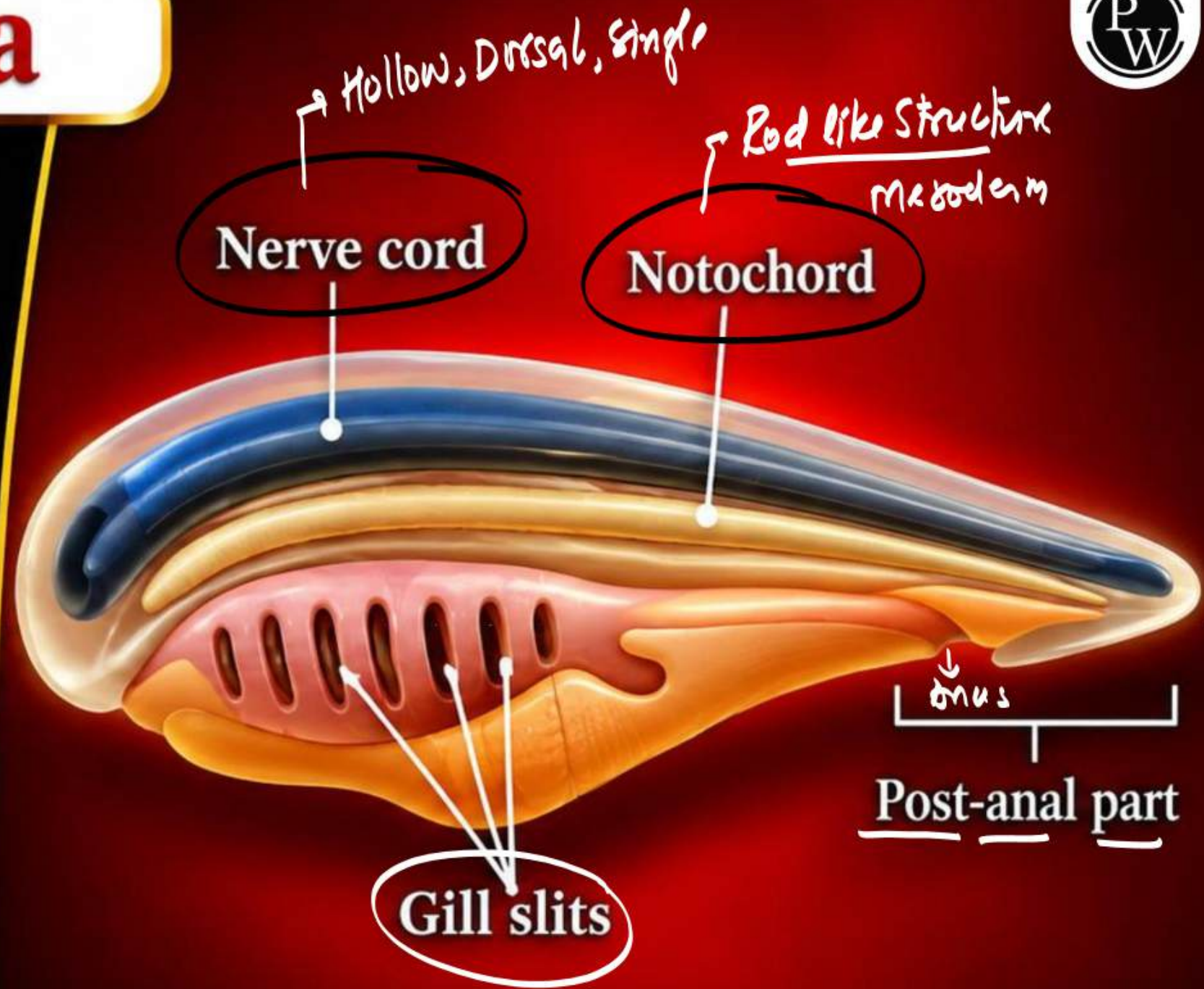
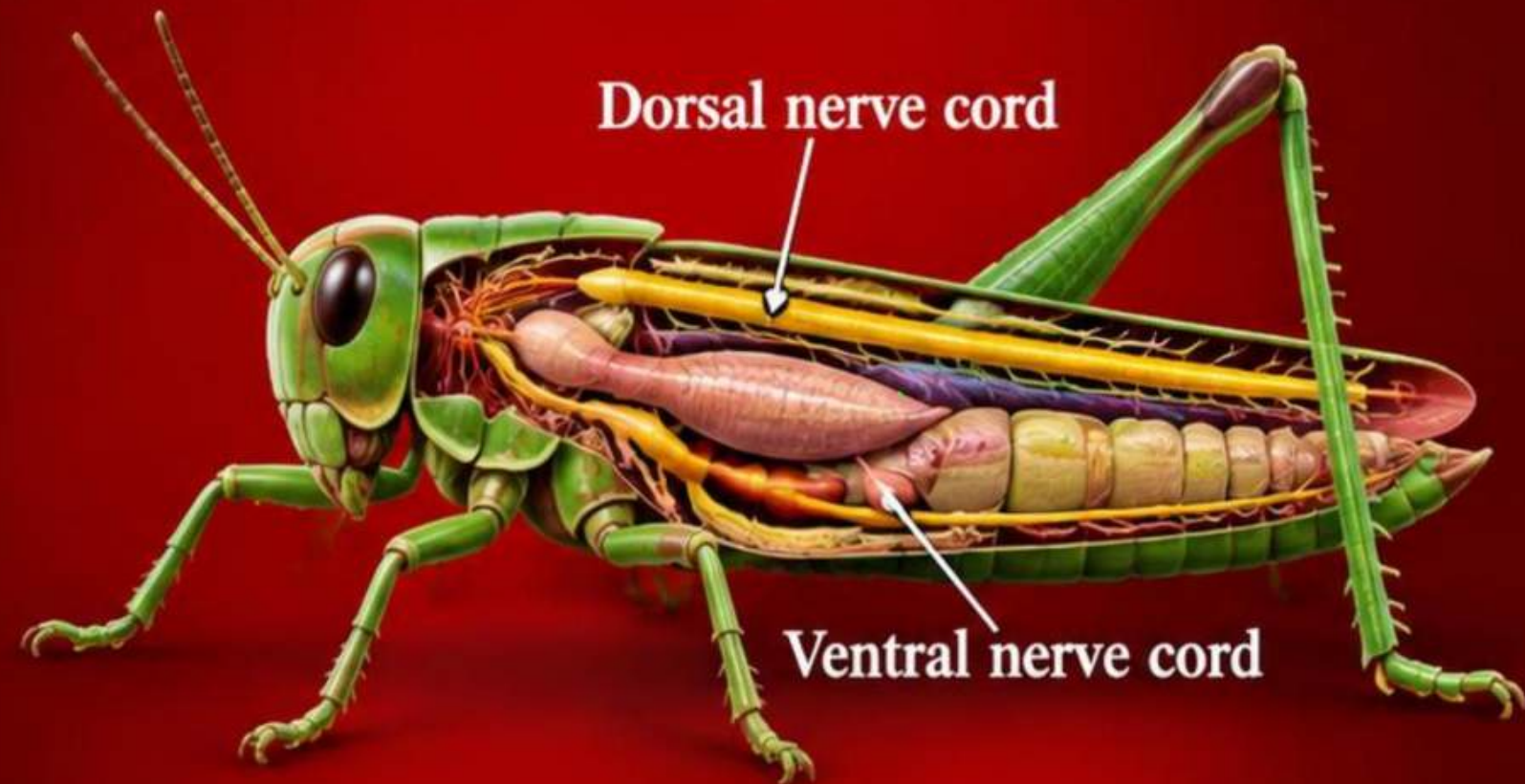
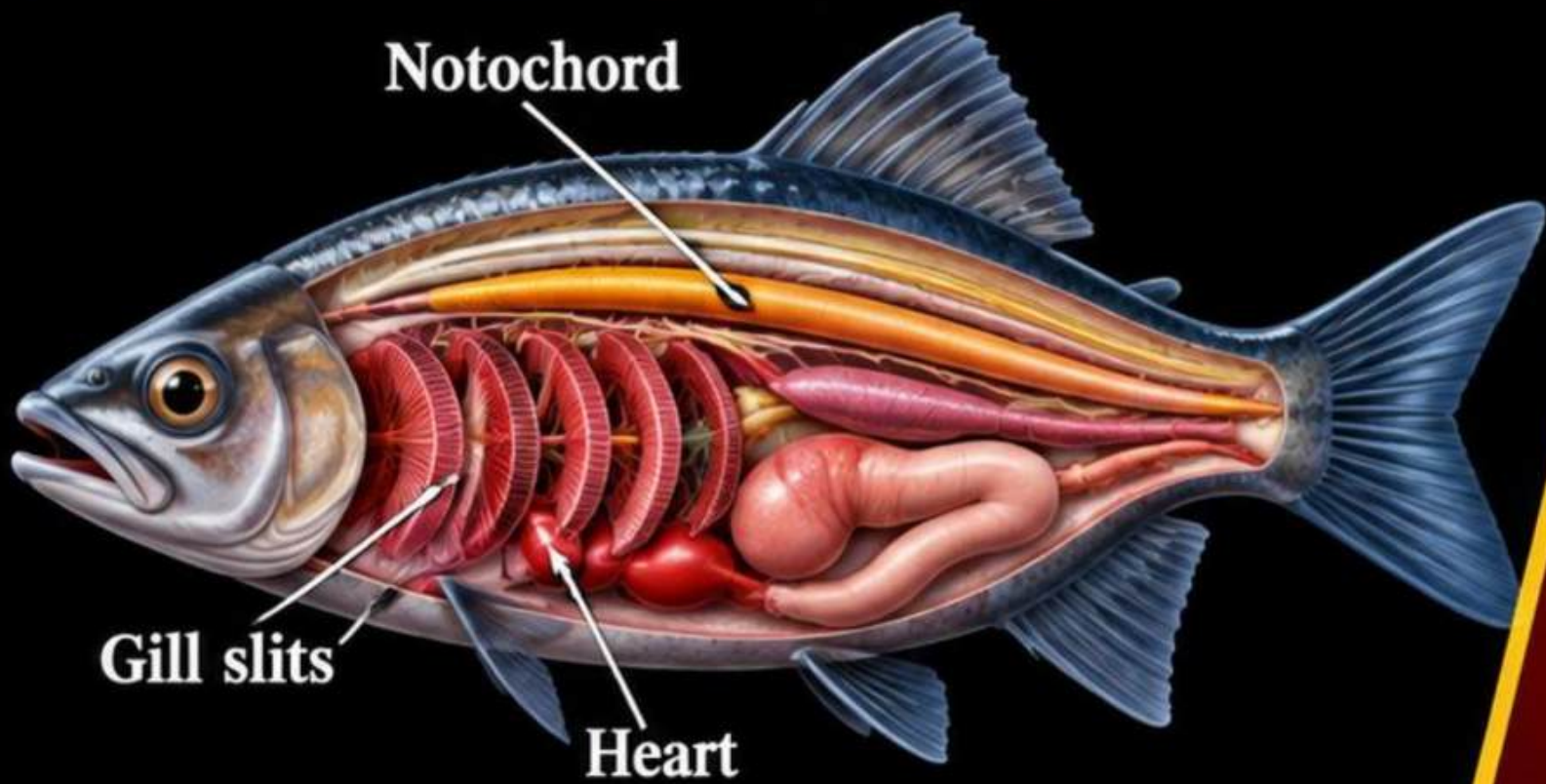
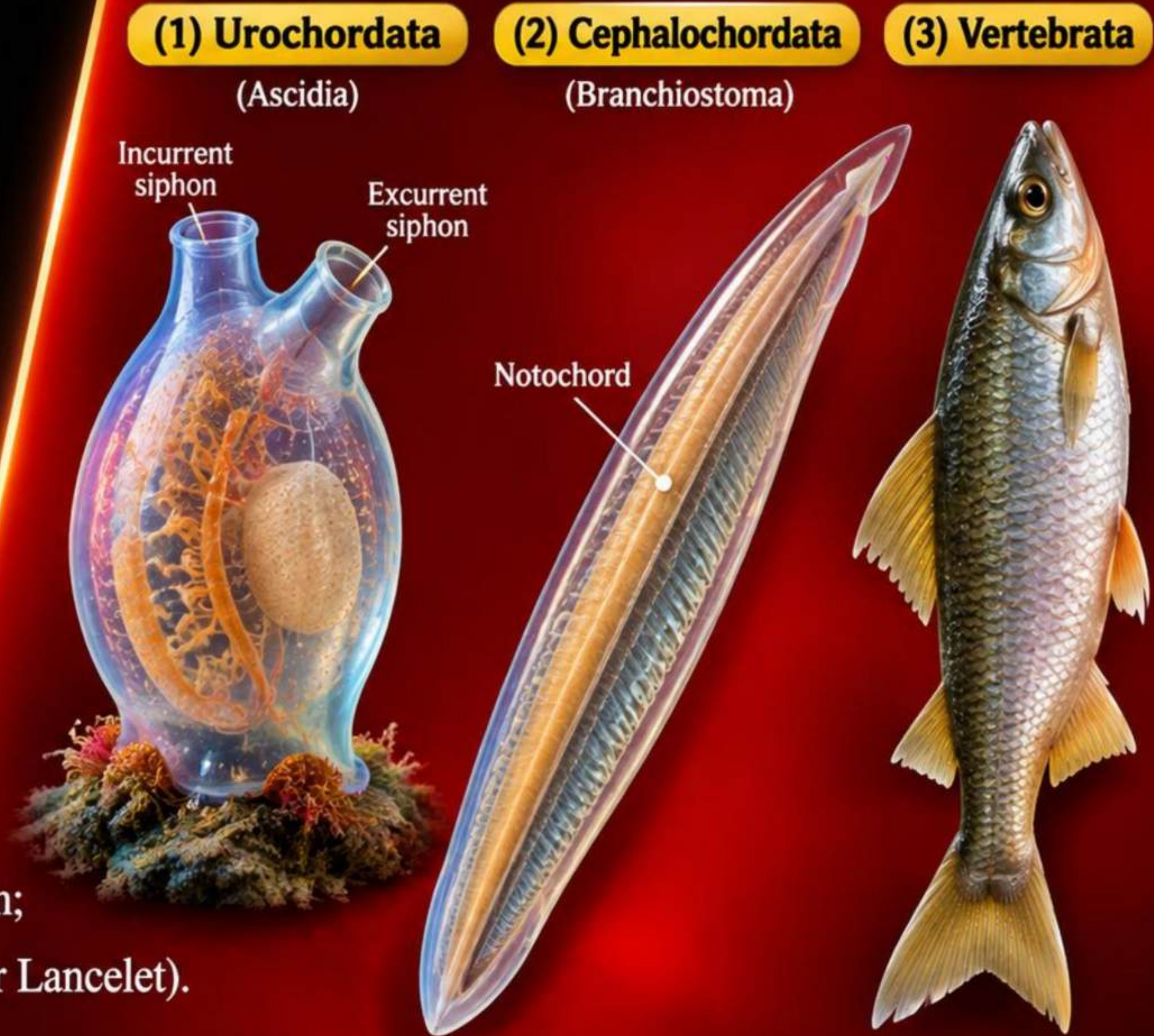


Figure 4.16 Chordata characteristics

S.No.	Chordates	Non-chordates
1.	Notochord present. ✓	Notochord absent. ✓
2.	Central nervous system is <u>dorsal</u> , <u>hollow</u> and <u>single</u> .	<u>Central</u> nervous <u>system</u> is <u>ventral</u> , <u>solid</u> and <u>double</u> .
3.	Pharynx perforated by gill slits. ✓	Gill slits are absent. ✗
4.	Heart is ventral. ✓	Heart is dorsal (if present). ✓
5.	A post-anal part (tail) is present. ✓	Post-anal tail is absent. ✗



- **Phylum Chordata** is divided into three subphyla:
- (1) Urochordata or Tunicata,
- (2) Cephalochordata and
- (3) Vertebrata
- Subphyla **Urochordata** and **Cephalochordata** are often referred to as protochordates and are exclusively marine.
- In **Urochordata**, notochord is present only in larval tail,
- while in **Cephalochordata**, ^{Notochord} it extends from head to tail region and is persistent throughout their life.
- **Examples:** Urochordata – Ascidia, Salpa, Doliolum;
- **Cephalochordata** – Branchiostoma (Amphioxus or Lancelet).



Chordal
no Cep

Sub-phylum

Vertebrata

(vertebral column)

Division

Agnatha
(lacks jaw)

Gnathostomata
(bears jaw)

Super Class

Pisces
(bear fins)

Tetrapoda
(bear limbs)

Class

1. Cyclostomata

Class

1. Chondrichthyes
2. Osteichthyes

Class

- ✓ 1. Amphibia
- ✓ 2. Reptilia
- ✓ 3. Aves
- ✓ 4. Mammals



Class – Cyclostomata

- All living members of the class Cyclostomata are ectoparasites on some fishes.
- They have an elongated body bearing 6-15 pairs of gill slits for respiration.
- Cyclostomes have a sucking and circular mouth without jaw.
- Their body is devoid of scales and paired fins.
- Cranium and vertebral column are cartilaginous.
- Circulation is of closed type.
- Cyclostomes are marine but migrate for spawning to fresh water.
- After spawning, within a few days, they die.
- Their larvae, after metamorphosis, return to the ocean.
Larva → Adult
- Examples: Petromyzon (Lamprey) and Myxine (Hagfish).

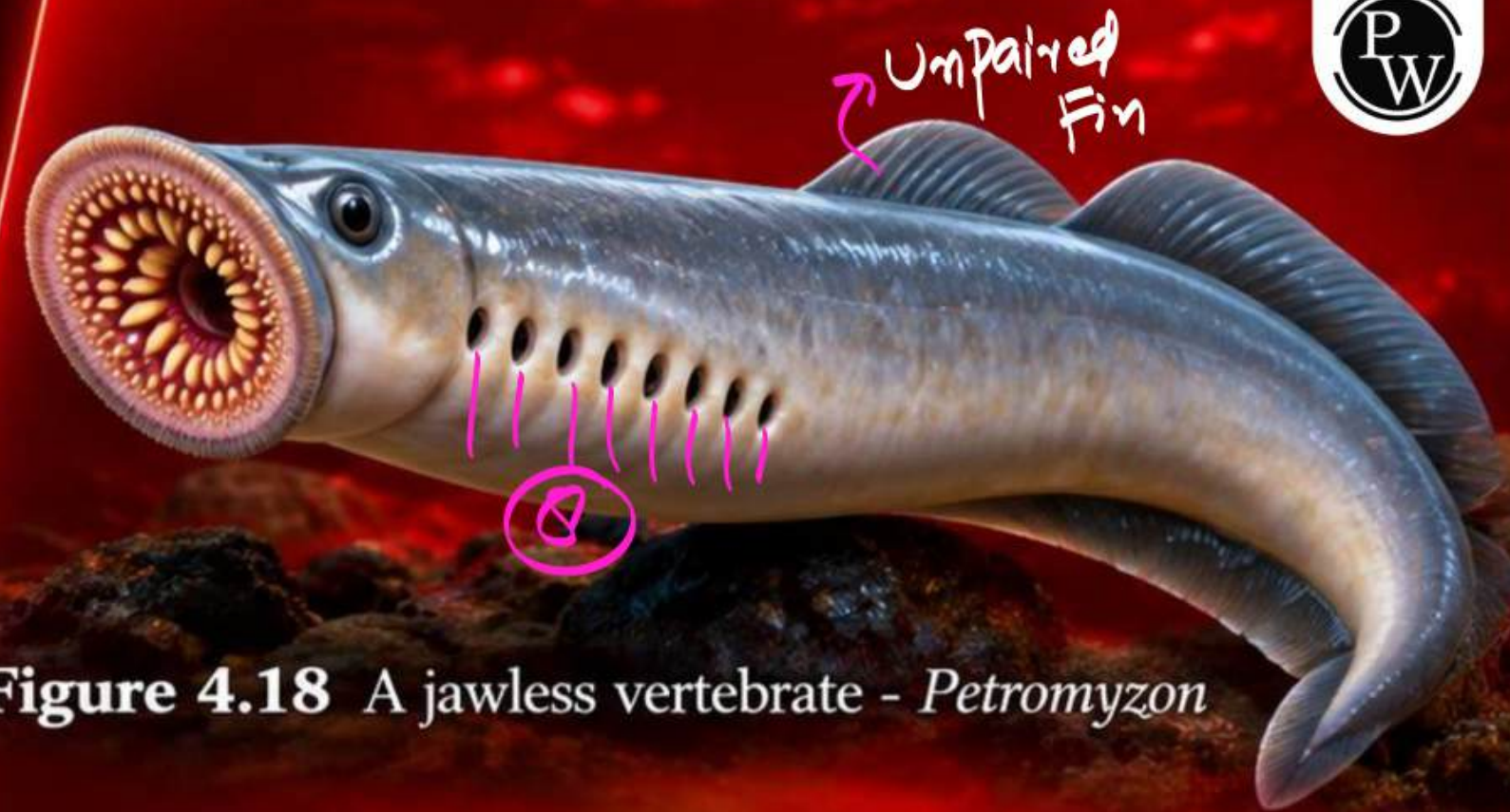


Figure 4.18 A jawless vertebrate - *Petromyzon*



Class – Chondrichthyes



- They are **marine animals** with ^{↓ cartilage} streamlined body and have **cartilaginous endoskeleton**.
- Mouth is located **ventrally**.
- **Notochord** is persistent throughout life.
- **Gill slits** are separate and without operculum (gill cover).
- The skin is tough, containing minute **placoid scales**.
- Teeth are modified **placoid scales** which are backwardly directed.
- Their jaws are **very powerful**.
- These animals are **predaceous**.
- Due to the **absence of air bladder**, they have to swim constantly to avoid sinking.



↓ Pelvic



- Heart is two-chambered (one auricle and one ventricle).
- Some of them have electric organs (e.g., Torpedo) and some possess poison sting (e.g., Trygon).
- They are cold-blooded (poikilothermous) animals, i.e., they lack the capacity to regulate their body temperature.
- Sexes are separate.
- In males pelvic fins bear claspers.
- They have internal fertilisation and many of them are viviparous.
- Examples: Scoliodon (Dog fish), Pristis (Saw fish), Carcharodon (Great white shark), Trygon (Sting ray)



Class – Osteichthyes

- It includes both marine and fresh water fishes with bony endoskeleton.
- Their body is streamlined.
- Mouth is mostly terminal.
- They have four pairs of gills which are covered by an operculum on each side.
- Skin is covered with cycloid/ctenoid scales.
- Air bladder is present which regulates buoyancy.
- Heart is twochambered (one auricle and one ventricle).
- They are cold-blooded animals.
- Sexes are separate.
- Fertilisation is usually external.
- They are mostly oviparous and development is direct.
- Examples: Marine – Exocoetus (Flying fish), Hippocampus (Sea horse);
Freshwater – Labeo (Rohu), Catla (Katla), Clarias (Magur);
Aquarium – Betta (Fighting fish), Pterophyllum (Angel fish).



Class – Amphibia

- Amphi : dual, bios, life, amphibians can live in aquatic as well as terrestrial habitats.
- Most of them have two pairs of limbs.
- Body is divisible into head and trunk.
- Tail may be present in some. (*Salamander*).
- The amphibian skin is moist (without scales).
- The eyes have eyelids.
- A tympanum represents the ear.
- Alimentary canal, urinary and reproductive tracts open into a common chamber called cloaca which opens to the exterior.
- Respiration is by gills, lungs and through skin.



- The heart is threechambered (two auricles and one ventricle).
- These are cold-blooded animals.
- Sexes are separate.
- Fertilisation is external. (water)
- They are oviparous and development is direct or indirect.
- Examples: Bufo (Toad), Rana (Frog), Hyla (Tree frog),
- Salamandra (Salamander), Ichthyophis (Limbless amphibia)



Class – Reptilia

- The class name refers to their creeping or crawling mode of locomotion (Latin, reperere or reptum, to creep or crawl).
- They are mostly terrestrial animals and their body is covered by dry and cornified skin, epidermal scales or scutes.
- They do not have external ear openings.
- Tympanum represents ear.
- Limbs, when present, are two pairs.
- Heart is usually three-chambered, but four-chambered in crocodiles.
- Reptiles are poikilotherms. (Cold Blooded)
- Snakes and lizards shed their scales as skin cast.



- Sexes are separate.
- Fertilisation is internal.
- They are oviparous and development is direct.
- Examples: Chelone (Turtle), ✓
- Testudo (Tortoise), ✓
- Chameleon (Tree lizard), ✓
- Calotes (Garden lizard), ✓
- Crocodilus (Crocodile), ✓
- Alligator (Alligator). ✓
- Hemidactylus (Wall lizard), ✓
- Poisonous snakes – Naja (Cobra), ✓
 Bangarus (Krait), Vipera (Viper). ✓



Class – Aves

- The characteristic features of Aves (birds) are the presence of feathers and most of them can fly except flightless birds (e.g., Ostrich).
- They possess beak.
- The forelimbs are modified into wings.
- The hind limbs generally have scales and are modified for walking, swimming or clasping the tree branches.
- Skin is dry without glands except the oil gland at the base of the tail.
- Endoskeleton is fully ossified (bony) and the long bones are hollow with air cavities (pneumatic).
- The digestive tract of birds has additional chambers, the crop and gizzard. (Crush).
- Heart is completely fourchambered.
- They are warm-blooded (homoiothermous) animals, i.e., they are able to maintain a constant body temperature.



- Respiration is by lungs.
- Air sacs connected to lungs supplement respiration.
- Sexes are separate. ✓
- Fertilisation is internal. ✓
- They are oviparous and development is direct. ✓
- Examples : Corvus (Crow), Columba (Pigeon), Psittacula (Parrot), Struthio (Ostrich), Pavo (Peacock), Aptenodytes (Penguin), Neophron (Vulture).

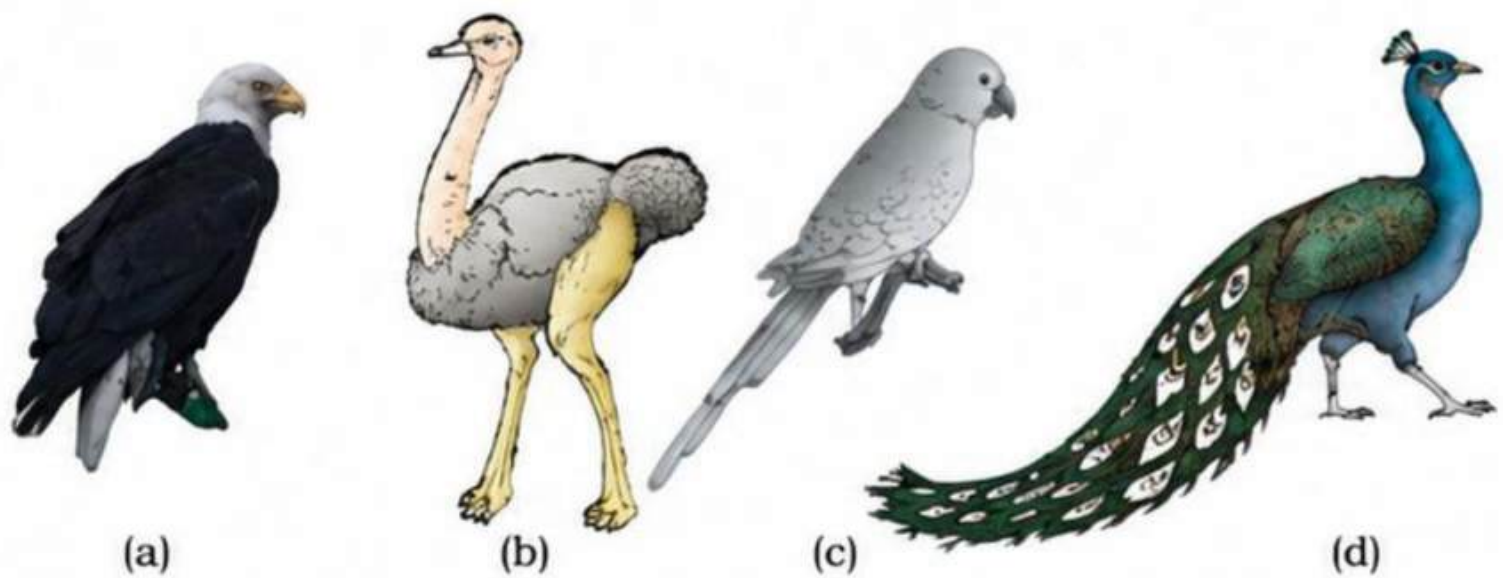


Figure 4.23 Some birds : (a) *Neophron* (b) *Struthio* (c) *Psittacula* (d) *Pavo*



Class – Mammalia

→ Mammary Glands

- They are found in a variety of habitats – polarice caps, deserts, mountains, forests, grasslands and dark caves.
- Some of them have adapted to fly or live in water.
- The most unique mammalian characteristic is the presence of milk producing glands (mammary glands) by which the young ones are nourished.
- They have two pairs of limbs, adapted for walking, running, climbing, burrowing, swimming or flying.
- The skin of mammals is unique in possessing hair.
- External ears or pinnae are present.
- Different types of teeth are present in the jaw.
- Heart is fourchambered. (Heterodonts).
- They are homoiothermous.
- Respiration is by lungs. Sexes are separate and fertilisation is internal.
- They are viviparous with few exceptions and development is direct.



Examples: Oviparous-Ornithorhynchus (Platypus);

- Viviparous - Macropus (Kangaroo),
- Pteropus (Flying fox),
- Camelus (Camel),
- Macaca (Monkey),
- Rattus (Rat),
- Canis (Dog),
- Felis (Cat),
- Elephas (Elephant),
- Equus (Horse),
- Delphinus (Common dolphin), Balaenoptera (Blue whale),
Panthera tigris (Tiger), Panthera leo (Lion).



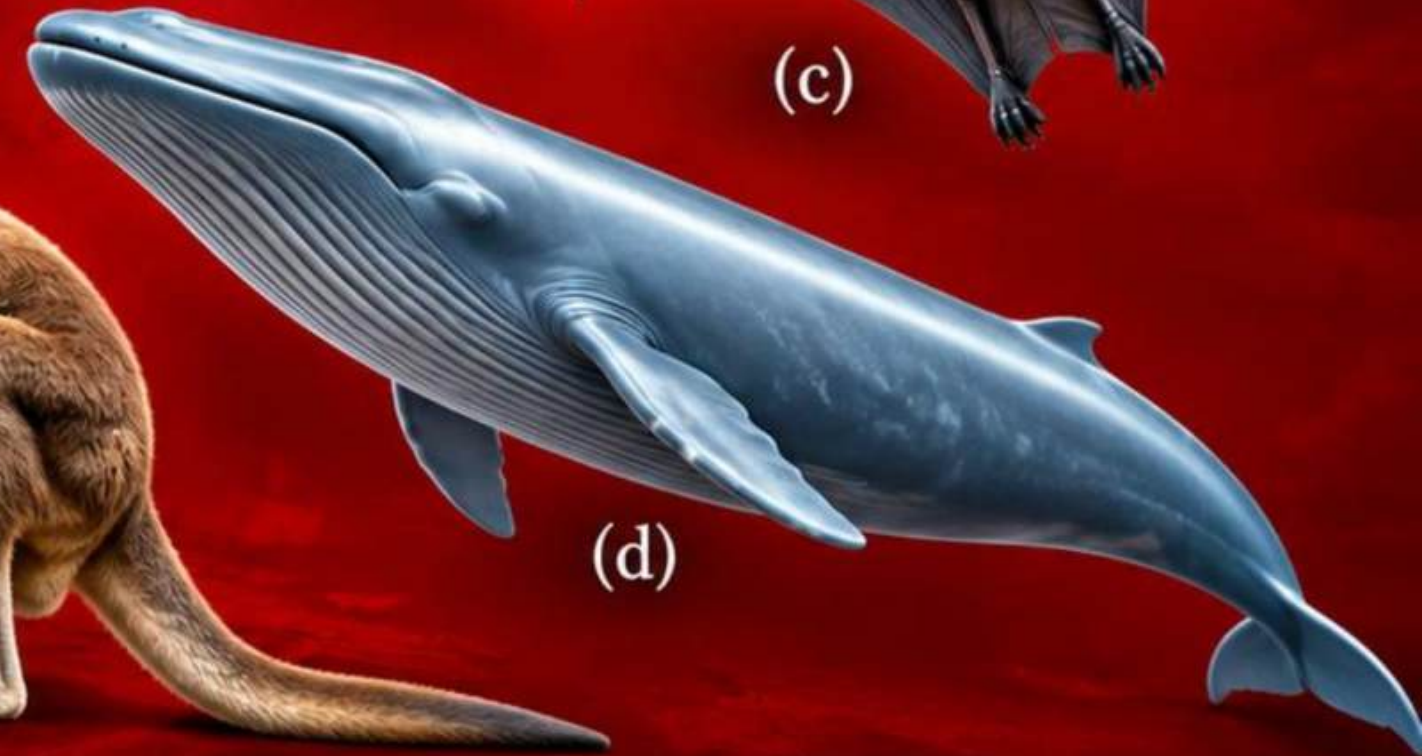
(a)



(c)



(b)



(d)

Figure 4.24 Some mammals : (a) *Ornithorhynchus* (b) *Macropus*
(c) *Pteropus* (d) *Balaenoptera*

HOME WORK

kE liye

Study

Practice

Better

Future

Good
Students
Good
Results 😊

- ✓ Plan
- ✓ Do
- ✓ Complete
- ✓ Repeat



Small
Steps
Big
Results



You
Can
Do It
!! 😊

Thank
You